Executive Brief — Customer Retention & Promotion Targeting

To: Marketing / CRM decision-maker | **Evidence:** UCI Online Retail II, 5,281 customers in the final 90-day temporal test beginning 2011-09-10.

Decision and main finding

Do not launch any of the three currently assumed discount campaigns. None offers positive simulated NET margin after fixed-unit product costs and contact expense. **No Promotion = £0 simulated incremental margin** is the decision reference. An available budget does not have to be spent.

The real-data baseline predicts repeat purchase with PR-AUC **0.750184** and ROC-AUC **0.789977** (actual repeat rate **43.36%**; naive PR-AUC **0.433630**). Future-gross-sales MAE is **£395.69**, RMSE **£2,016.14**. These are temporal-test prediction metrics, not evidence that promotion causes additional sales.

Advanced modeling strengthens but does not change the decision: GBM classification improves final-test PR-AUC to **0.769056**, while regression accuracy is temporally mixed across five secondary origins. The baseline remains the interpretable business reference; monitoring is more appropriate than changing promotion assumptions.

Policy comparison — simulated NET margin

Scenario	Non-targeted mean	Random-policy 95% interval	RFM	Proposed / No Promotion
Conservative	-£17,417.59	[-£21,707.94, -£14,023.76]	-£55,333.45	£0 / £0
Moderate	-£37,407.57	[-£46,583.42, -£30,081.21]	-£118,314.35	£0 / £0
Aggressive	-£90,677.15	[-£112,076.51, -£73,360.59]	-£277,865.97	£0 / £0

Non-targeted averages **1,000 seeded random campaigns**, each contacting **1,500** customers; its interval describes variation across random customer selections, **not a causal confidence interval**. RFM also contacts 1,500. Both spend **£750** at **£0.50/contact**. Proposed selects only positive-NET-margin customers, up to the same capacity; it selects zero and spends £0 in all three scenarios. Source: `outputs/scenario_table.csv`.

Assumptions and target groups

- Discounts/base sales-volume growth: **5%/10%, 10%/20%, 20%/35%**. Gross margin:

40%. Growth is an assumed proportional increase, not conversion probability.

- Segment growth multipliers: Champions 0.5, Loyal 1.0, At-risk 1.3, Lost 0.4;

these are business assumptions, not measured treatment responsiveness.

- Unit product costs remain fixed. NET contribution is

`v*((1+r)*(g-d)-g) - £0.50`, with nonnegative predicted gross sales `v`.

- **Nurture:** positive NET contribution, ranked under the budget. None currently

qualify. **Retain:** positive-value Champions/Loyal without discounted offers; any service/relationship benefit is unmeasured. **No-offer:** remaining customers.

Sensitivity and decision gate

At 10% discount, positive-value At-risk customers need median assumed base growth of **26.75%**, exceeding the configured 20%; their median supportable discount at 20% growth is **7.91%**. Explore smaller discounts or independently supported growth assumptions rather than changing assumptions merely to produce profit.

First profitable points on the tested discount × growth grid are **5%/20%** (£7,175.92), **10%/30%** (£1,424.27), and **15%/50%** (£962.39). No campaign is profitable at 20% discount through 50% growth. In separate one-at-a-time checks, 35% growth at 10% discount gives £3,990.80. These are exploratory, non-causal scenarios—not evidence to launch a campaign. Validate any changed assumption through a randomized pilot before rollout.

Limitations

There is no randomized treatment/control: no causal uplift, true promotion ROI, or actual incremental profit is claimed. Forecasts describe historical gross positive sales, not untreated revenue, net returns or lifetime CLV. Of 1,067,371 raw rows, 34,335 exact duplicates are retained under the documented source policy. Outliers, return exclusions, time drift and uncertain growth assumptions limit transfer to current customers. Detailed methodology and provenance are in the model card and `outputs/run_manifest.json`; the current evaluation score artifact is `data/processed/customer_test_scores.csv`. Removing 34,335 exact duplicate rows in a separate sensitivity leaves 5,281 customers and the same no-campaign recommendation; it does not establish that duplicates are errors.